

# Prarthana Krishnamurthy

[krishnamurthy.prarth@gmail.com](mailto:krishnamurthy.prarth@gmail.com) | [github/pkrishna1801](https://github.com/pkrishna1801) | [linkedin/prarthana181k](https://www.linkedin.com/in/prarthana181k) | [pkrishna1801.github.io](https://pkrishna1801.github.io) | 857-396-4456

## Summary

Machine Learning Engineer with 4+ years of experience building predictive and generative models alongside the production data pipelines that feed them. Delivers models end to end: PyTorch and scikit-learn training, FastAPI and MCP serving, PySpark ETL, and Docker with CI/CD deployment.

## Experience

### Machine Learning Intern

May 2026 – Jul 2026

Alkermes Inc.

Waltham, MA

- Built supervised models (PyTorch, scikit-learn) predicting Chemical properties at  $R^2$  0.7 and Spearman 0.73
- Deployed a FastAPI serving layer for the trained models, cutting inference latency 2.5x
- Exposed the model endpoints through an MCP server, letting AI agents invoke them as callable tools
- Established the team's ML deployment infrastructure (Docker, GitLab CI/CD), enabling reliable model releases
- Applied SHAP to translate black-box model outputs into explanations for non-technical stakeholders

### Machine Learning Research Assistant

Jan 2025 – Apr 2026

Northeastern University (Advisor: Prof. Dino Konstantopoulos)

Boston, MA

- Architected transformer and diffusion models using flow matching for non-autoregressive text generation
- Processed 30GB+ raw text corpora with spaCy into training-ready NLP datasets for model development
- Cut DataLoader memory by extracting frozen BERT embeddings outside autograd, eliminating OOM at large batch sizes
- Scaled training across 8x NVIDIA H100 GPUs via PyTorch FSDP, sustaining 95%+ GPU utilization
- Resolved sampling error accumulation by implementing DPM-Solver++, reducing discretization error to 0.65
- Delivered SOTA metrics including BLEU-4: 0.267 and ROUGE-L: 0.742, demonstrating end-to-end ownership

### Software Engineer

Sep 2021 – Dec 2023

Societe Generale Global Solution Centre

Bangalore, KA

- Architected an AWS data lake supporting 1M+ daily records, designing ingestion and partitioning strategies
- Productionized 3+ Dockerized ML applications on Azure Kubernetes Service, processing 1M+ daily transactions
- Improved fraud detection precision 18% at fixed recall with transaction-velocity and NetworkX graph features
- Built data drift detection and automated retraining triggers on transaction feature distributions, reducing model degradation incidents by 30% across quarterly retraining cycles
- Engineered PySpark pipelines for raw-to-silver processing, enforcing schemas and deduplicating records

### Software Engineering Intern

Mar 2021 – May 2021

Capgemini

Bangalore, KA

- Built 15+ RESTful microservices (Java 17, Spring Boot), cutting API latency 35% with JPA caching
- Implemented multi-tenant microservices, ensuring high availability and data consistency for enterprise clients
- Engineered unit and integration test suites (JUnit, Mockito) to catch regressions and ensure API reliability
- Developed a client-facing React UI over those microservices, exposing services and data directly to clients

## Publications

- [Non-Autoregressive Text Generation Using Diffusion Models](#), Master's Thesis, Northeastern University

## Technical Skills

**Programming Languages:** Python (NumPy, SciPy, Pandas, Matplotlib), SQL, Java, React, Bash, Linux

**ML/DL Frameworks:** PyTorch (FSDP, DDP), TensorFlow, Keras, scikit-learn, XGBoost, LightGBM

**Deep Learning & Generative AI:** Transformers, Diffusion Models, Flow Matching, Optuna, LLMs, RAG

**Cloud, MLOps & Deployment:** AWS (S3, ECS, Lambda, Kinesis, DynamoDB, CloudWatch), Azure (Data Lake, AKS), Docker, Kubernetes, Terraform, FastAPI, MCP, REST APIs, Microservices, Git, CI/CD

**Data Engineering:** Airflow, PySpark, Elasticsearch, ETL/ELT Pipelines, Parquet, Kafka, Apache Flink, Hadoop

**Mathematics & Statistics:** Linear Algebra, Calculus, Probability, Optimization, Information Theory, SDE/ODE Solvers

## Education

Northeastern University, MS in Computer Software Engineering (GPA: 3.84/4.0)

Jan 2024 – Dec 2025

Visvesvaraya Technological University, BE in Electronics and Communication

Aug 2017 – Aug 2021